

***IBM Professional
Certification Program***

Exam C1000-207

STUDY GUIDE SERIES

IBM watsonx Orchestrate AI Engineer v1 - Associate



Purpose of Exam Objectives

When an exam is developed, Subject Matter Experts work together to define the role the certified individual will fill. They define the tasks and knowledge that an individual would need to successfully perform this job role done the product or solution. This creates the foundation for the objectives and measurement criteria, which form the basis of the certification exam. Question writers then use these objectives to develop exam questions.

It is recommended that you review these objectives and ask yourself the following questions:

- Do you know how to complete the task in the objective?
- Do you know why that task needs to be done?
- Do you know what will happen if you do it incorrectly?

If you are not familiar with a task, go through the objective, perform that task in your own environment and read more information on the task. Use the Study Resource reference links below and search for the topic of the task. Review the documentation that you find until you *can* answer 'yes' to the above questions. If there is an objective on a task, there is a high likelihood that you **WILL** see a question about it on the actual exam. Review the recommended learning designed to prepare you to take the certification exam.

After reviewing the objectives in this guide and completing your own research, take the assessment exam. While the assessment exam does not indicate which specific questions were answered incorrectly, it does indicate overall performance by section. This is a good indicator of preparedness or if further preparation is warranted.

Study Resources

Below is a high-level list of resources to help when you are preparing for the certification exam. This list is not exhaustive and is meant to help you should you need more information on the topics listed below in the Study Guide. Use the high-level reference links and search for the topic of the task.

Reference Links:

IBM watsonx Orchestrate Documentation

<https://www.ibm.com/docs/en/watsonx/watson-orchestrate>

IBM Community (watsonx and AI blogs, articles, discussions)

<https://community.ibm.com/community/user>

IBM Developer Portal

<https://developer.ibm.com>

IBM Developer APIs Catalog

<https://developer.ibm.com/apis>

IBM Developer Articles

<https://developer.ibm.com/articles>

watsonx Orchestrate Developer Documentation

<https://developer.watson-orchestrate.ibm.com>

watsonx Orchestrate Security & Governance Portal

<https://connect.watson-orchestrate.ibm.com>

Detailed Exam Objectives

Section 1: Platform Architecture and Core Concepts

This section accounts for approximately 13% of the exam.

TASK: 1.1 Understand the core capabilities of the watsonx Orchestrate platform

1. Understand AI agents
 - 1.1. Describe AI orchestration
 - 1.2. Explain the business need for AI Agent orchestration
2. Describe Software Hub
 - 2.1. Articulate the interdependence of Software Hub and Orchestrate
3. Define the watsonx Orchestrate platform and its purpose within the automation ecosystem.
4. Identify and describe each core capability:
 - 4.1. Environments (Dev, Test, Prod)
 - 4.2. Agents and Assistants
 - 4.3. Tools and Connections
 - 4.4. Agentic Workflows
 - 4.5. Knowledge Bases
 - 4.6. ADK (Agent Development Kit)
 - 4.7. Catalog of Reusable Assets
 - 4.8. AI Gateway and model policies
5. Explain how these capabilities work together to enable end-to-end automation.
6. Recognize where governance, scalability, and security apply within the platform.
7. Differentiate between built-in and custom (ADK-based) components.

TASK: 1.2 Scope watsonx Orchestrate for a business use case

1. Identify when watsonx Orchestrate is the appropriate solution
2. Characteristics of processes suitable for Orchestrate
(repetitive, rule-based, multi-step, standardized inputs, predictable outcomes)
3. Scenarios where Orchestrate provides value
(task automation, human-in-the-loop workflows, cross-system orchestration)
4. Scenarios that are NOT suitable
(highly custom development, complex ML pipelines, unstructured decision-making)
5. Analyze an existing business process to prepare it for automation
 - 5.1. Identify process triggers, actors, systems, and required data
 - 5.2. Recognize dependencies and handoffs
 - 5.3. Determine required decisions, approvals, and conditions
 - 5.4. Identify desired outcomes and success metrics

Section 2: Agent and Assistant Integration

This section accounts for approximately 20% of the exam.

TASK: 2.1 Design and implement integrations

1. Define and distinguish Assistants and Agents
2. Summarize the capabilities of Assistants and Agents
3. Identify the specific use cases and tasks that such integration needs to support
4. Implement integration in WXO
 - 4.1. Describe the steps that you need to go through for such integration

TASK 2.2: Configure external agents

1. Describe external agents
2. Describe the business value of external agents
3. Distinguish AI assistants, Agents, collaborator agents and external agents
4. Configure External A2A agents by using ADK
5. Demonstrate how to configure and integrate different (external) agents
6. Identify/List agents supported by watsonx Orchestrate

TASK 2.3: Create Multi-Agent Orchestration

1. Define Multi-Agent Orchestration
2. Describe Agent Catalog and its business value
3. Describe the business value of Multi-Agent Orchestration

4. Demonstrate how to apply Multi-Agent Orchestration in W XO
 - 4.1. Agents from third party platforms
 - 4.2. Agents from watsonx.ai
 - 4.3. W XO assistants as Collaborator Agents
 - 4.4. AI assistants as Agents
 - 4.5. External A2A agents by using ADK

TASK 2.4: Describe Langflow integration as a visual orchestration option

1. Describe Langflow and its role in watsonx Orchestrate
2. Describe the business value of Langflow integration
3. Distinguish Langflow tools from AI assistants, Agentic Workflows, tools, collaborator agents, and external agents
4. Configure Langflow tools for watsonx Orchestrate using ADK Start ADK
5. Integrate Langflow tools into watsonx Orchestrate agents
6. Identify/List supported Langflow components and integration capabilities

Section 3: Workflow and Orchestration Design

This section accounts for approximately 14% of the exam.

TASK 3.1 Build Agentic Workflows

1. Describe how to create an agentic workflow
2. Explain how to add user activities to a workflow
3. Articulate adding generative prompts in a workflow
4. Explain how code blocks are used
5. Define the flow outline used in Orchestrate
 - 5.1. Trigger → sequence of actions → decision points → outcomes
 - 5.2. Identify required agents and skills

TASK 3.2 Integrate with MCP tools

1. Describe Model Context Protocol (MCP) and its role in AI agent architecture
2. Explain the business value of MCP integration in watsonx Orchestrate
3. Distinguish MCP tools from other tool integration approaches
4. Register tools with MCP server and validate functionality using MCP Inspector.
5. Import MCP tools into watsonx Orchestrate using ADK
6. Create and configure agents on watsonx Orchestrate that consume MCP tools
7. Expose and consume tools through ContextForge MCP Gateway

TASK 3.3: Pre-built domain agents

1. Describe pre-built domain agents and their role in watsonx Orchestrate
2. Explain the business value of pre-built domain agents
3. Identify categories of pre-built domain agents available in watsonx Orchestrate
4. Explore the watsonx Orchestrate Agent Catalog
5. Configure secure connections for pre-built agents to your enterprise applications
6. Customize and deploy pre-built domain agents
7. Describe real-world scenarios:
 - 7.1. Demonstrate multi-agent orchestration for HR workflows, and Sales prospecting workflows.

Section 4: Agent Development

This section accounts for approximately 18% of the exam.

TASK: 4.1 Design Agents

1. Define an agent's purpose, scope, and expected output.
2. Identify required tools, knowledge bases, connections, and data dependencies
3. Decide when an agent should call tools vs retrieve from knowledge base vs. Rely on LLM reasoning
4. Design logic flows (decision points, input/output handling).
5. Apply UX and conversational design principles (tone, clarity, accessibility).
6. Identify required agents and tools
7. Embed governance, ethics, and tone guidelines into agent design.
8. Evaluate the design for reusability and scalability.
9. Break down a business workflow into automation steps
 - 9.1. Sequence tasks in logical order
 - 9.2. Determine which parts belong to:
 - 9.2.1. an agent
 - 9.2.2. a tool
 - 9.2.3. a connection, or
 - 9.2.4. a flow branch
 - 9.3. Understand entry points and exit points for automation
 - 9.4. Identify steps requiring human approval or review

TASK: 4.2 Build Agents

1. Explain the overall process of building an agent
 - 1.1. Main stages from setup through readiness for testing
 - 1.2. Key inputs and outputs of the build process
2. Identify the structural elements that make up an agent
 - 2.1. Tools, connections, metadata, knowledge bases, and Agentic workflows
 - 2.2. How these elements interact to deliver a complete capability
3. Describe the purpose and use of the Agent Development Kit (ADK)
 - 3.1. Extending platform functionality through custom components
 - 3.2. When to use the ADK versus existing catalog assets
4. Define agent configuration elements
 - 4.1. Metadata such as name, description, owner, and permissions
 - 4.2. Environment selection and policy considerations
5. Use tools and connections in Agents
 - 5.1. Tools as discrete actions or composite capabilities
 - 5.2. Tools as containers for grouped functionality
 - 5.3. Connections as mechanisms linking to external systems
6. Explain integration and connectivity concepts
 - 6.1. Purpose of authentication, credentials, and endpoints
 - 6.2. How data moves between external sources and the agent
7. Summarize readiness activities before agents move to testing
 - 7.1. Verifying that all assets and connections are defined
 - 7.2. Ensuring documentation and version details are complete

TASK: 4.3 Test Agents

1. Understand the evaluation framework and process
2. Understand evaluation datasets
 - 1.1. Record user stories
 - 1.2. Generate user stories
3. Understand agent evaluation metrics
4. Understand how testing ties into Observability
5. Test for vulnerabilities

TASK: 4.4 Understand Agentic Prompting

1. Explain the concept and purpose of agentic prompting in AI systems.
2. Identify common prompting techniques used when designing AI agents.
3. Explain methods used to evaluate and test prompts for agent performance.
4. Describe security risks associated with prompts in AI agents.
5. Identify security controls used to mitigate prompt-based attacks.

TASK 4.5: Connect agents to channels

1. Explain the purpose of connecting AI agents to communication channels
2. Identify supported channel types for AI agent interactions
3. Describe the process for configuring an agent to connect to a channel
4. Explain security considerations when connecting agents to external channels
5. Describe how to test and validate agent interactions through channels

Section 5: Model Management

This section accounts for approximately 14% of the exam.

TASK: 5.1 Develop Agents With Orchestrate Copilot

1. Install and configure the Orchestrate Copilot for Visual Studio Code
 - 1.1. Initialize an agent development workspace
2. Understand copilot commands

TASK: 5.2 Model Selection

1. Choose a LLM
 - 1.1. Understand model appropriateness for agent task type
 - 1.2. Understand cost, accuracy, and latency tradeoffs
 - 1.3. Consider input types (input modalities)
 - 1.4. Understand third-party models
 - 1.4.1. Describe which third-party models are available

TASK 5.3: Integrate with the AI Gateway

1. Define AI Gateway
 - 1.1. Describe the business value of AI Gateway
 - 1.2. Describe the features of AI Gateway
2. Describe how AI Gateway enables multi-model and multi-provider Orchestration for agents and workflows
3. Describe example use cases that AI Gateway supports
4. List the deployment models for AI Gateway
5. Summarize the capabilities of AI Gateway

TASK: 5.4 Configure Model Policies

1. Identify different types of model policies

2. Understand how to create, edit, update, and remove model policies
3. Understand when to apply specific model policies
 - 3.1. Understand how model policies determine selection and failover among LLM models
 - 3.2. Understand how agent ops and governance dashboards lead to model policies
 - 3.3. Apply guardrails through model policies
4. Understand how to import existing model policies for reuse across environments

Section 6: Security, Compliance, and Observability

This section accounts for approximately 11% of the exam.

TASK: 6.1 Authenticate, authorize and audit

1. Understand the authentication mechanisms used in watsonx Orchestrate
2. Understand how role-based access control (RBAC) governs user permissions and task execution within watsonx Orchestrate
3. Learn how authentication and authorization are applied across Orchestrate services
4. Understand how to configure secure connections for Orchestrate APIs and gateways
5. Learn how to enable and interpret activity and audit logs
6. Understand the security checklist provided for authentication, authorization, and audit compliance
7. Understand how to manage secure connections and integration
8. Understand when to place authentication, authorization, and audit concerns in context

TASK: 6.2 Monitoring, logging, and observability

1. Understand how to enable and access audit logs in watsonx Orchestrate
2. Understand how to enable and configure external logging
3. Understand how to monitor agent analytics via the built-in dashboards
4. Understand how observability is supported and how to connect to external observability tools
5. Understand how to interpret key observability metrics to diagnose performance and reliability issues
6. Understand how to monitor LLMs with Langfuse

TASK: 6.3 Governance and compliance

1. Understand how watsonx Orchestrate provides built-in governance dashboards, policy controls and monitoring of AI agents and workflows

2. Understand how compliance controls are embedded
3. Understand how enterprise regulatory frameworks is supported via policy enforcement and visibility
4. Understand how governance activities are applied across the lifecycle of AI agents, assistants, and workflows
5. Understand how watsonx Orchestrate aligns with enterprise security and compliance frameworks
6. Understand how to configure features for Security and Privacy by Design (SPbD)
7. Understand the considerations related to governance and compliance when designing the agent's behavioral profile

Section 7: Deployment, Scaling, Optimization, and Resiliency

This section accounts for approximately 10% of the exam.

TASK 7.1: Build and Manage Environments

1. Create and configure new watsonx Orchestrate environments - Set up environment instances with appropriate naming conventions, regions, and resource tiers
2. Manage environment variables and configurations - Define and update environment-specific settings, API keys, and connection parameters
3. Implement environment isolation - separation between dev/test/prod
4. Monitor environment health and resource utilization - Track performance metrics, capacity limits, and system alerts
5. Perform environment backup, restore, and migration - Export/import configurations and manage environment versioning

TASK 7.2: Build and Manage Knowledge Bases

1. Create and configure knowledge bases - Initialize knowledge base instances with appropriate schemas, storage backends, and retrieval configurations
2. Ingest and process documents - Upload documents in various formats (PDF, HTML, JSON), manage chunking strategies, and handle metadata
3. Implement vector embeddings and indexing - Configure embedding models, vector databases, and indexing parameters for semantic search
4. Manage knowledge base versioning and updates - Track document versions, perform incremental updates, and manage snapshot/rollback procedures
5. Optimize retrieval performance and relevance - Tune search parameters, implement filtering, and monitor query latency and accuracy metrics

6. Implement knowledge base security and access control - Configure document-level permissions and data isolation

TASK 7.3: Manage Agents, Assistants, and Gateways

1. Deploy and configure conversational assistants - Create assistant instances, connect to knowledge bases, and configure conversation flows
2. Build and manage autonomous agents - Define agent goals, connect skills/tools, implement decision-making logic, and configure agent behavior
3. Configure and manage API gateways - Set up gateway endpoints, implement routing rules, configure rate limiting and authentication
4. Connect and orchestrate skills - Integrate pre-built and custom skills, manage skill dependencies, and configure skill chaining
5. Implement versioning and A/B testing - Deploy multiple assistant/agent versions, route traffic for testing, and manage rollback procedures
6. Monitor agent/assistant performance - Track conversation metrics, error rates, user satisfaction, and response times
7. Troubleshoot and optimize agent behavior - Debug conversation flows, optimize prompt engineering, and improve response accuracy

TASK 7.4: Implement CI/CD Pipelines

1. Design CI/CD pipeline architecture - Define pipeline stages (build, test, deploy), identify artifacts, and establish promotion workflows across environments
2. Implement version control integration - Connect Git repositories, manage branching strategies, and track configuration changes
3. Automate artifact building and packaging - Create automated processes for packaging skills, assistants, and configuration bundles
4. Implement automated testing - Configure unit tests, integration tests, and conversation flow validation in pipeline stages
5. Configure deployment automation - Set up automated deployment triggers, approval gates, and rollback mechanisms
6. Integrate with DevOps tools - Connect with Jenkins, GitLab CI, Tekton, or other pipeline tools and configure watsonx.orchestrate CLI/APIs
7. Implement monitoring and alerting in pipelines - Configure pipeline health checks, deployment verification tests, and notification systems
8. Manage secrets and credentials in pipelines - Securely handle API keys, tokens, and sensitive configuration data

Next Steps

1. Take the sample test for [IBM watsonx Orchestrate AI Engineer v1 – Associate](#)
2. Take the assessment test for [IBM watsonx Orchestrate AI Engineer v1 - Associate](#)

If you pass the assessment exam, visit <https://home.pearsonvue.com/ibm> to schedule your testing sessions.

If you failed the assessment exam, review how you did by section. Focus attention on the sections where you need improvement. Keep in mind that you can take the assessment exam as many times as you would like (\$30 per exam); however, you will still receive the same questions only in a different order.